

THE NEXT DECADE IN TECHNOLOGY

2019	2020	2021	2022	2023
 James Webb Space Telescope launched The James Webb Space Telescope (JWST) is a next generation space telescope to replace the Hubble.	 5G standard is released 5G is expected to be commercially available from 2020, designed for peak speeds of 20 gigabits per second.	 Male birth control is available The first male birth control options are expected to be available by 2021, as the researchers move from testing to commercial deployment.	 Fusion reactor demonstrated Humans are expected to generate electricity from nuclear fusion in 2022, and then will begin away from coal based and fission power.	 Supersonic air travel makes a comeback There have been no supersonic passenger flights since the Concorde stopped operations in 1993, Boom hopes to start them again in 2023.
 Computers break the exaflop barrier The exaflop barrier is expected to be broken by an Intel machine, developed in collaboration with European researchers.	 3D printed human organs used for drug testing The first organs, made up of 3D printed tissue, is expected to be available to pharmaceutical companies by 2020, for drug testing.	 82 percent of web traffic is video The bandwidth hungry nature of video content means that video will continue to dominate web traffic.	 Driverless hover taxis start operating in Dubai The autonomous air taxi service with a range of 50 kilometers will have finished five years of testing, and be ready for regular operations.	 20 Billion connected IoT devices According to forecasts by Ericsson, there will be 20 billion IoT devices by 2023, with 1.8 billion having their own cellular connections.
 Bionic Eye implants are commercially available Pixium Vision, Second Sight and Retina Implant are all in various stages of developing and testing eye implants to tackle visual ailments.	 Screens are ultra thin OLEDs Most of the screens in all displays, from smartphones to televisions, are expected to be made up of flexible, ultra thin OLEDs.	 We will be able to eat lab grown meat Meat has been cultured in labs for years, but it has not yet been cheap enough to introduce it at scale.	 Nanotech clothing There will be environmental concerns, but nanotech will be common in clothing, to make it stain resistant, waterproof, and antimicrobial.	 Driverless high speed trains in France Testing for the drone train is expected to begin in 2019, and France hopes to regularly run the autonomous trains from 2023.
 Galileo Navigation System operational Galileo is a global navigation system being created by the European Union through the European Space Agency.	 Level 5 autonomous self driving cars The first fully autonomous cars, where the occupants only need to feed in a destination, is expected to launch in 2020.	 First Indian manned space mission ISRO had plans for a manned mission in an orbiting module with three crew members on board.	 China completes its space station The Chinese large modular space station is the culmination of the Tiangong project, which will establish a space station in low earth orbit.	 OSIRIS-REx returns with samples from Benu The first sample return mission from an asteroid is expected to return to Earth by 2023, around the same time private companies intend to start mining space rocks.
 Apple Electric Car The company is known for its secrecy, so expect little details about "Project Titan", ahead of an official announcement. A number of engineers have been working on this.	 Exploration of the red planet 2020 is a big year for Mars exploration, with a number of launches, including pathfinding missions to identify spots for manned expeditions.	 Translation earbuds work as expected We have these now, but by 2021 machine translation is expected to be good enough to allow real time conversations between people who do not know a common language.	 Portable hyperimager Humans are predicted to have devices capable of providing "superhuman vision", capable of seeing in various wavelengths, and through fog or dust.	 80 percent of the population is online Electricity and connectivity continues to be a problem.